

1.

International Islamic University Chittagong
Center for General Education (CGED)

Semester End Examination -Spring 2026

Course Code: GEEL-1106

Course Title: Advanced English

Time: 2 hours & 30 minutes

Marks: 45+05 (Listening)

Reading part 25

The history of man is the history of war. Throughout the ages, man has been concerned with the problem of preventing war. If in the past, nations had not wanted to go to war with one another, no association of nations would have been necessary to outlaw war. But history has proved to mankind that the nations of the world have not been willing to observe these conditions.

The League of Nations, the first association of nations established to work for peace, was founded in 1919. For four years from 1914, war had raged throughout Europe. The fighting in this war had been more destructive than anything that mankind had ever experienced. The League of Nations aimed at outlawing war and settling international disputes by peaceful means—by discussion instead of by force. For twenty-five years the League of Nations struggled to survive. With the outbreak of the Second World War, in 1939, it ceased to function. If it had been able to keep its promise of enforcing disarmament, there would not have been another war. Britain and France had disarmed, but other nations had not followed their example. Some nations had defied the League of Nations and gone to war with their neighbors.

These events led to the Second World War in 1939. This war raged over the continents and seas of the world from 1939 to 1945. Millions of soldiers, sailors and airmen were killed. Thousands of innocent civilians were the victims of deadly weapons. If there had been no war, all this suffering could have been avoided. When the war ended, the people of all the nations began praying for a secure, peaceful world without any fear of war. This desire for world peace led to the founding of the United Nations Organization (UNO).

On October 24 1945, representatives of fifty-one nations met to form an association called the United Nations Organization. The two main aims of UNO are the maintenance of international peace and security and the promotion of human welfare throughout the world. Since then the organization has survived with difficulty. It has faced a series of international disputes that could have involved the world in a nuclear war. The successful handling of these disputes has proved that UNO can help in the maintenance of world peace. If UNO had failed to settle them, the world might by now have been involved in a global war and completely destroyed. If it can continue to settle disputes peacefully, the fear of a worldwide disaster will disappear. If it fails, there may be no further hope for mankind.

1. Answer the following questions as instructed

13

- a. Both world organizations were similarly successful. Is the statement right or wrong?
- b. *To declare something unlawful*- find a synonym for this expression from the passage.
- c. Why did the League of Nations fail to achieve its aim?
- d. *Struggled to survive*- Make a sentence of your own with the expression.
- e. How many nations sent representatives to the first meeting of UNO?
- f. He is working hard to get _____ in his job. Fill in the blank with a word from the passage.
- g. *World-wide disaster*- Make a sentence of your own with the expression.
- h. It had been able to keep its *promise of enforcing disarmament*. Rewrite the sentence replacing the words in italics.
- i. The orphan boy maintains his life with _____. Fill in the blank with a word from the passage.
- j. What makes the people hopeful about the League of Nations?

- k. *Innocent civilians*- Make a sentence of your own with the expression.
 l. *To look after*- find a synonym for this expression from the passage.
 m. The word *suffering* underlined in the passage is gerund or participle.

In the face of mounting withdrawal pressure, the central bank yesterday dissolved the bank's entire board and appointed an administrator. But much of the liquidity damage had already been done. The country's largest private-sector bank, which had begun to recover under the interim government after years of massive loan irregularities and financial misconduct linked to the S Alam Group, is now under renewed liquidity pressure as deposit outflows accelerate.

The bank's current account with the central bank was in a positive position until May 31. But within the past two weeks, it has moved into a deficit of about Tk 4,000 crore, show data. A current account with the central bank is a master reserve account held by commercial banks and financial institutions. It acts as the clearing mechanism for interbank transactions and allows institutions to settle daily obligations.

When the balance falls into deficit, banks can face difficulties in fund transfers and clearing operations, which can disrupt services for customers. Md Altaf Hossain, managing director (current charge), acknowledged the pressure. He told The Daily Star yesterday that the deficit had widened due to the bank's reliance on liquidity support from the central bank. As a result, the bank has imposed a Tk 50,000 cash withdrawal limit per customer at the branch level, he added. However, some customers said branches in Dhaka were allowing withdrawals of only Tk 10,000 amid the ongoing pressure. The Bangladesh Bank yesterday injected Tk 2,500 crore in special liquidity support to help the country's largest shariah-based lender address the acute cash shortage. Bangladesh Bank Executive Director and Spokesperson Arif Hossain Khan confirmed the development to The Daily Star.

According to internal data, Islami Bank's total deposits fell from Tk 1,84,382 crore on May 31 to Tk 1,80,141 crore on June 7. Facing the pressure, the lender sought Tk 10,000 crore in liquidity assistance from the Bangladesh Bank last week.

On May 24, just as the country was preparing for a week-long Eid holiday, the Bangladesh Bank appointed Md Khurshid Alam, a former deputy governor, as chairman of Islami Bank hours after the previous chairman resigned. The appointment came as a surprise and triggered immediate concern among stakeholders.

2. Answer the following questions as instructed:

12

- IBBL PLC is one of the largest private banks in the country. Is the statement right or wrong?
- Interrupt*- find a synonym for this expression from the passage.
- What do you think the cause of liquidity crises at IBBL PLC?
- Ongoing pressure*- Make a sentence of your own with the expression.
- What event initiated and triggered the problem in the bank?
- He had a _____ attack in his heart. Fill in the blank with a word from the passage.
- Daily obligations* - Make a sentence of your own with the expression.
- Md Altaf Hossain, the MD *acknowledged the pressure*. Rewrite the sentence replacing the words in italics.
- The man has _____ on others to maintain himself. Fill in the blank with a word from the passage.
- Acute cash shortage* - Make a sentence of your own with the expression.

- k. *To generate*- find a synonym for this expression from the passage.
 l. Is the word **Facing**, underlined in the passage, gerund or participle?

Grammar part -10

3. Answer the following questions as instructed.

- The students were given reception by the Mayor who got scholarship. (Correct the sentence)
- I was ill. He heard the news. He came back from journey early. His purpose was to see me. (Combine these sentences to a simple sentence)
- The students who are indifferent _____ their regular studies are sure to fail. (Use preposition in the gap).
- None can catch a falling star. (falling is here used as gerund or present participle?)
- Despite _____, he was sentenced to death. (Use words or phrase to fill up the gap).
- He does not like apples ----does he like oranges. (Use appropriate conjunction).
- It is high time Fariha _____. (Use words to complete the sentence).
- Do all the students know what have they to do now? (Correct this sentence)
- Our sons are brave. They laid down their lives for the cause of our motherland. The name of our motherland is Bangladesh. We remember them with respect. (Combine these sentences to a complex sentence)
- If I had had enough money, I (establish) a hospital like Ad-deen Hospital for the poor. (Use correct form of the bracketed verb).

Writing part -10

- It is said that chat GPT has turned into a potential threat to genuine talent of human being. Do you agree? Write an argumentative essay on this topic explaining your own position. 05
- Write a letter to the Manager of Walton Plaza, Agrabad appreciating the products that you have bought recently? 05

N.B. Listening Test will be taken by the course teacher at a convenient time.

05

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
Semester Final Examination

Program: B. Sc. in CSE
 Course Code: MATH-1107
 Time: 2:30 hours



Semester: Spring 2026
 Course Title: Mathematics-I
 Total Marks: 50

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Please answer the several parts of a question sequentially.
 (iii) Separate answer script must be used for separate groups.
 (iv) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1:	Compute the functions, limit and continuity of a function, derivatives, integrals and extrema of single-variable and/or multivariable functions.
CLO2:	Understand the techniques of differentiation and integration.
CLO3:	Demonstrate the applications of differentiation and integration.

Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group - A

		Marks	CLO	DL
1.	a) Define partial derivatives. If $u = \ln(x^2 + y^2)$ then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$	5	CLO1	R&U
	Or) Define Homogeneous function. If $u = \sin^{-1} \frac{x+y}{\sqrt{x}+\sqrt{y}}$ then using the Euler's theorem on homogeneous function show that, $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{\tan u}{2}$			
	b) Define maxima and minima of a function. Find the maxima and minima of the function, $y = 2x^3 - 9x^2 + 12x - 3$	5	CLO1	U
2.	a) A proton moves in an electric field such that its acceleration (in cms^{-2}) is $a = -\frac{20}{(1+2t)^2}$; where t is in seconds. Find the velocity as a function of time if $v = 30 \text{ cms}^{-1}$ when $t = 0$.	5	CLO2	U
	Or) Evaluate the Integral, $\int \frac{3x dx}{x^2 - x - 2}$			
	b) (i) Evaluate the Integral, $\int x \tan^{-1} x dx$	3	CLO2	U
	(ii) Evaluate the Integral, $\int e^x (2x + x^2) dx$	2	CLO2	U

Group - B

- | | | | | | |
|----|-----|--|---|------|---|
| 3. | a) | Evaluate the Integral, $\int_0^1 x^3 dx$ by geometrically. | 5 | CLO2 | U |
| | b) | If $I_n = \int_0^{\pi/4} \tan^n \theta d\theta$ and $n > 1$ then using the reduction formula prove that, $n(I_{n-1} + I_{n+1}) = 1$ | 5 | CLO2 | U |
| | Or) | Using the properties of definite integral show that,
$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi^2}{4}$ | | | |
| 4. | a) | Evaluate the triple integral, $\int_0^1 \int_0^x \int_0^y x^3 y^3 z dz dy dx$ | 5 | CLO2 | U |
| | b) | Using Gamma and Beta function prove that, $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ | 5 | CLO2 | U |
| | Or) | Using Gamma and Beta function Prove that $\int_0^{2\pi} \cos^4 x dx = \frac{3\pi}{4}$ | | | |
| 5. | a) | Find the circumference of the circle, $x^2 + y^2 = a^2$ about the y-axis. | 4 | CLO3 | A |
| | b) | Find the area of the surface generated by revolving the arc of the curve of $y = x^3$ from $x = 0$ to $x = 1$ about the x-axis. | 3 | CLO3 | A |
| | c) | Find the volume of the solid of revolution generated by rotating the curve $y = \sqrt{x}$ between $y = 0$ to $y = 4$ about the y-axis. | 3 | CLO3 | A |

International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering
Semester Final Examination, Spring-2026
Course Code: PHY-1101 Course Title: Physics-I
Time: 2 hours 30 minutes Full Marks: 50

Answer all the questions. Separate answer scripts must be used for Group-A and Group-B. The figures in the right-hand margin indicate full marks. Course Learning Outcomes and Bloom's Levels are mentioned in additional columns.

Group-A

- | | | | |
|---|------|---|---|
| 1. a) What is the Doppler effect of sound? | CLO1 | R | 1 |
| b) Find an expression of a stationary wave using two opposite progressive waves of the same frequency and amplitude. Hence, investigate the conditions for nodes and antinodes. | CLO2 | U | 6 |

Or

Derive an expression for the average kinetic energy of a particle undergoing SHM.

- | | | | |
|---|------|---|---|
| c) Two trains, A and B, are approaching each other, and their velocities are 108 km/h and 144 km/h, respectively. The frequency of a note emitted by A as heard by the passengers in B is 1170 Hz. Calculate the frequency of the note as heard by the passengers in A. The velocity of sound is 350 m/s. | CLO3 | A | 3 |
|---|------|---|---|

Or

A 0.500-kg cube connected to a light spring for which the force constant is 20.0 N/m oscillates on a horizontal, frictionless track. Given that the amplitude of the motion is 3.00 cm. Calculate the (a) Total energy of the system and (b) the maximum speed of the cube.

- | | | | |
|--|------|---|---|
| 2. a) Describe resonance with an example. | CLO1 | R | 2 |
| b) Derive the differential equation for a particle oscillating under Simple Harmonic Motion. | CLO2 | U | 5 |
| c) A particle performs simple harmonic motion (SHM) according to the equation, | CLO3 | A | 3 |

$$x = 5 \sin \left(4t + \frac{\pi}{6} \right),$$

where x is in cm, and t is in seconds.

Find: (i) Amplitude, (ii) Initial Phase (epoch), (iii) Time Period, (iv) Frequency, (v) Phase difference between two positions of the particle 0.5 s apart, (vi) Displacement after t = 1.5s.

Group-B

- | | | | |
|--|------|---|---|
| 3. a) Deduce an expression for the change in entropy of an ideal Carnot heat engine in terms of temperatures of the hot and cold reservoirs. | CLO2 | U | 7 |
|--|------|---|---|

Or

Develop an expression for the intensity pattern due to single-slit Fraunhofer diffraction.

- | | | | |
|--|------|---|---|
| b) Find the efficiency of a Carnot engine working between the steam point and ice point. | CLO3 | A | 3 |
| 4. a) Derive the expression for the intensity at a point due to the interference of light using Young's double-slit experiment, and hence find conditions for consecutive dark and bright fringes. | CLO2 | U | 7 |
| b) A light source emits light of two wavelengths, 4500 Å and 6000 Å. The source is used in a Young's double-slit experiment. The distance between the slits and the screen is 2.0 m, and the slit separation is 0.04 mm. Calculate the separation between the 4th-order bright fringes due to the two wavelengths. | CLO3 | A | 3 |

Or,

A gas is compressed adiabatically from $V_1 = 1.0 \text{ m}^3$ to $V_2 = 0.02 \text{ m}^3$. The initial pressure is $2.5 \times 10^5 \text{ Pa}$, and final pressure is $6.5 \times 10^5 \text{ Pa}$. If $\gamma = 1.67$, calculate the work done.

- | | | | |
|---|------|---|---|
| 5. a) State and explain the law of Malus. | CLO1 | R | 2 |
| b) Show that when light is incident on a transparent material at the Brewster angle, the reflected and refracted rays are mutually perpendicular to each other. | CLO2 | U | 5 |
| c) Given that the transmission axis of an analyzer makes an angle of 30 degrees with the plane of vibration of the plane-polarized light. Find the intensity of light after the analyzer. | CLO3 | A | 3 |

Final Examination, Spring-2026

Course Code: **EEE-1121**

Time: 2 hour 30 minutes

Program: B.Sc. Engineering in CSE

Course Title: Basic Electrical Engineering

Full Marks: 50

[Answer all the questions from the following. Figures in the right margin indicate full marks]

Course Outcomes (COs) and Blooms Levels are mentioned in additional columns.

Part-A

- 1) a) Compare the RMS value and the average value of a signal. Find out the RMS and average value of the given signal shown in Fig-1a. CO-1 Ap 05

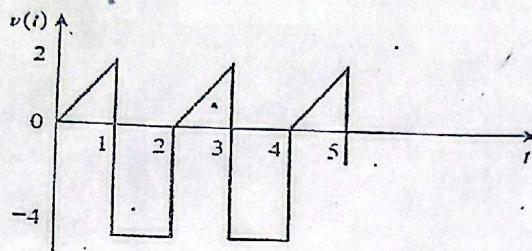


Fig-1a

OR

- a) Consider a sine wave signal with an peak to peak 10 volts, varying from 0V to 5V over a cycle. Calculate the average and RMS values of this sine wave signal. CO-1 Ap 05

- b) Two signals of Current and Voltage of a circuit is given below: CO-3 An 05
 $i = 2 \cos(\omega t + 10^\circ)$
 $v = 3 \sin(\omega t - 30^\circ)$

Find the phasor relationship between these two signals. Additionally, create a neat waveform sketch for both signals to visually represent their behavior.

- a) Determine the instantaneous power equation for an AC circuit, where the voltage and current equations are given by: CO-3 An 05
 $v(t) = 80 \cos(10t + 20^\circ)$
 $i(t) = 15 \sin(10t + 60^\circ)$

- b) In the circuit of Fig-2b, calculate the average power absorbed by the resistor and capacitor. Find the average power supplied by the voltage source. CO-3 Ap 05

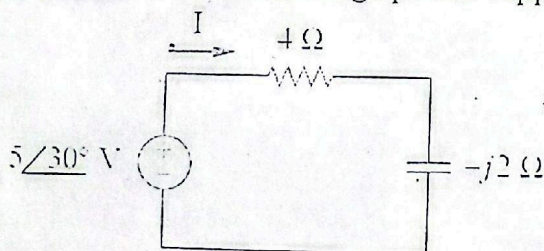


Fig-2b

OR

- b) Find the value of the capacitance needed to correct a load of 140 kVAR at 0.85 lagging pf to unity pf. The load is supplied by a 110 Volt (rms), 60 Hz line. CO-3 Ap 05

Part-B

- 3) a) Explain the phasor relationship between voltage and current in purely inductive and purely capacitive AC circuits. Derive the mathematical expressions and illustrate the phase relationship using phasor diagrams. CO-1 Ap 05

OR

- 3) a) Find the voltage $v(t)$ in a circuit described by the integrodifferential equation. CO-3 Ap 05

$$\frac{dv}{dt} + 5v(t) + 4 \int v dt = 20 \sin(4t + 10^\circ) \text{ V}$$

- 3) b) For the following pairs of voltages and currents, determine whether the load involved in Fig-2 is a Capacitor, an inductor, or a resistor. Also, calculate out the value of C, L, or R if sufficient information is given. CO-3 Ap 05

a. $v=50 \cos(157t+20^\circ)$, $i = 5 \sin(157t+110^\circ)$

b. $v=10 \sin(377t+10^\circ)$, $i = 5 \sin(377t-80^\circ)$

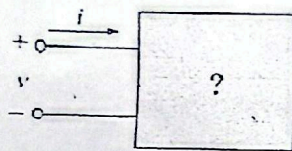


Fig-3b

- a) What is filter in electrical circuits? Briefly discuss about the bandpass filter. CO-3 Ap 05

- b) Given that $R=20 \text{ k}\Omega$ and $C=1200 \text{ pF}$ in a Low Pass Filter. CO-3 An 05

a. Sketch the normalized plot for voltage gain A_v .

b. Determine the cut-off frequency.

c. Determine the magnitude of voltage gain A_v . $f = \frac{1}{2\pi RC}$

- a) Define 3-phase AC circuit and draw the all four possible balanced 3-phase connections. CO-3 Ap 05

- b) A balanced Δ -connected load having an impedance $10 - 8j \Omega$ is connected to a Δ -connected positive-sequence generator having $V_{ab} = 330 \angle 0^\circ$. Calculate the phase currents of the load. CO-3 Ap 05

OR

- b) A Y-connected source with $V_{an}=100 \angle 10^\circ \text{ V}$ is connected to a Y-connected load with an impedance of $8+j4 \Omega$ per phase. Determine the line currents using single-phase analysis. CO-3 Ap 05

International Islamic University Chittagong

Centre for General Education (CGED)

Semester End Examination, Spring -2026

Course Code: GEEM-1101

Course Title: Text of Ethics and Morality

Faculty of Science

Marks: 50

Duration: 2 hours 30 minutes

Instructions:

i. All Questions are compulsory

ii. Figures in the right margin indicate full marks.

iii. Course Learning Outcome (CLO) and Bloom's levels are mentioned in additional columns.

Bloom's Level of the Questions							
Letter of Symbol	R	U	App	An	E	C	
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create	
SL	Text of the Questions				Marks	CLOs	Bloom's Level
01	a) What are the objectives and learning outcomes of this course? b) Who will be the inheritors of Jannatul Firdaus?				5 X 2= 10	CLO-2	R
02	a) Elucidate the values of youth mentioning the story of surah al kahaf and Surah Maryam. b) Discuss the Quranic verses that are related to modern science.				5 X 2= 10	CLO-2	U
03	a) What steps do you think should be taken to make society free from drugs and gambling? Explain. B) "Allah has permitted trade and has forbidden interest" Explain the verse.				5 X 2= 10	CLO-2	C
04	a) Illustrate the religious and political rights in Islam mentioning the Quranic verses. b) Explain the rights of divorcee as mentioned in the holy Qur'an. Or a) Analyze the importance of acquiring knowledge as mentioned in the holy Quran. b) Explain the lessons of surah al-Asr.				5 X 2= 10	CLO-3	C & An
05	a) Evaluate the necessity of Luqman's sermon for the family- peace. b) Who are ideal servants? Illustrate their features in the light of surah Furqaan.				5 X 2= 10	CLO-3	C & E

International Islamic University Chittagong
 Department of Computer Science & Engineering
 B.Sc. in CSE, Final Term Examination, Spring 2026
Course Code: CSE-1121 Course Title: Computer Programming I
Total Marks: 30 Time: 90 minutes

[Answer *all* the questions. Figures in the right-hand margin indicate full marks.
 Separate answer scripts must be used for Group A and Group B]

CO
DL

Group - A

1/a) What is the output of the following C code snippet? Explain.

3 CO2
C2

```

int main() {
    int i,j;
    for (i = 1; i <= 4; i++) {
        for (j = 1; j <= 4; j++) {
            if (i == j)
                continue;
            if (i * j > 6)
                break;
            printf("%d ", i + j);
        }
        printf("\n");
    }
    return 0;
}
    
```

b) Suppose you used ChatGPT to get a code to print the **exact pattern** given below and got the code given in the code section below. Unfortunately, you ran the code on your local machine and found that the output does not match the expected output. Now modify the code so that you can get the exact output.

3 CO4
C4

Expected Pattern	Code
<pre> 1 1 2 1 2 3 1 2 1 </pre>	<pre> int i, j; for(i=1;i<=3;i++){ for(j=1;j<=3;j++){ printf("%d \n",j); } printf(" "); } for(i=2;i>=1;i--){ for(j=1;j<=3;j++){ printf("%d \n",j); } printf(" "); } </pre>

c) A **Disarium Number** is a number defined such that the sum of its digits raised to the power of their respective positions is equal to the number itself. The position indexing starts from 1 for the leftmost digit.

4 CO2
C3

Sample Input	Sample Output
135	Disarium Number
80	Not Disarium Number

Explanation for Sample 1: $135 \rightarrow 1^1 + 3^2 + 5^3 = 1 + 9 + 125 = 135$.

[OR]

A security system generates a password score by adding the factorial of each digit in a numeric password. Write a C program to calculate the Password Score of a given numeric password.

Sample Input	Sample Output
253	128
401	5065

Explanation Sample 1: If the password is 253. $2! = 2$, $5! = 120$, $3! = 6$
So, Password Score = $2 + 120 + 6 = 128$

a) Determine the output of the following program and explain the role of the static variable.

```
int x, y;
void update() {
    static int y = 5;
    x = x + y;
    y--;
    printf("y = %d, x = %d\n", y, x);
}
int main() {
    int i;
    for(i = 0; i < 2; i++)
        update();
    return 0;
}
```

b) Write a function prototype and its complete definition for a function named **findMaxMark**. The function should take an integer array (representing the marks of N students) and its size N as arguments, that returns the maximum mark in the array.

[OR]

Write a function named **countEvenDigits** that receives an integer as its only parameter and returns the number of even digits in the number. Print the returned value in the **main()** function.

Sample Input	Sample Output
248135	3

c) Determine the output of the following recursive function. [Here, YZ means the last two digits of your Student ID i.e. if your ID is C261010, then 9YZ means 910]

```
void fun(int a){
    if(a == 0){
        printf("\n");
        return;
    }
    printf("%d ", a%10);
    fun(a/10);
    printf("%d ", a%10);
}
int main(){
    int n = 9YZ;
    fun(n);
    return 0;
}
```

Group - B

a) Write a C program to declare a one-dimensional integer array of size 6 named arr. Take the elements as input from the user and print them in reverse order. 3 CO2
C2

b) This snippet is designed to "delete" all occurrences of the number 0 by shifting subsequent elements to the left. However, if there are two consecutive zeros in the array, it fails to remove the second one. Identify the error(s) and write the corrected code. 3 CO4
C4

```
int arr[6] = {1, 0, 0, 2, 3, 0};
int n = 6;
for (int i = 0; i < n; i++) {
    if (arr[i] == 0) {
        for (int j = i; j < n - 1; j++) {
            arr[j] = arr[j + 1];
        }
        n--;
    }
}
```

c) Write a C program to read two integers N, M ($N, M > 1$), then read a 2D array of size $N \times M$. The program must check whether all four corners of the matrix contain the exact same value. Print "Equal Corners" if they match, and "Not Equal" otherwise. 4 CO2
C3

Sample Input	Sample Output
<pre>3 3 5 0 5 0 1 0 5 0 5</pre>	Equal Corners
<pre>2 4 1 2 3 4 1 2 3 4</pre>	Not Equal

[OR]

Write a C program that takes a 2D array of integers A of size $N \times M$ as input. Calculate and print the average of each row. Also, print the overall average of all the elements in the matrix.

Sample Input	Sample Output
<pre>3 3 1 2 3 4 5 6 7 8 9</pre>	<pre>Average of row 1: 2.00 Average of row 2: 5.00 Average of row 3: 8.00 Overall average: 5.00</pre>

4 a) Write a custom C function named `xstrcpy` that mimics the behavior of `strcpy()` from `<string.h>`. It takes two character arrays as parameters and copy the contents of the second into the first. 3 CO2
C3

b) Explain the differences between *pass by value* and *pass by reference* in C with suitable examples. 3 CO3
C2

c) Write a C program to input a string and check whether it is a palindrome or not. A string is called a palindrome if it reads the same forward and backward. Print "Palindrome" if it is, otherwise print "Not Palindrome". Also print the length of the string. 4 CO2
C3

Sample Input	Sample Output
madam	Palindrome and length of string: 5
sir	Not Palindrome and length of string: 3

[OR]

You are given a string S containing a student's full name, all in lowercase, with single spaces between words (e.g., "md kamrul hasan"). Write a C program to capitalize the first letter of every word in the string and print the updated string.

Sample Input	Sample Output
abdur rahman tareq	Abdur Rahman Tareq

5 a) Define a C structure named WeatherReading that contains:

- dayOfMonth (integer)
- windDirection (string for directions like "North", "South-West")
- humidity (integer)
- rainfall (float)

3 CO3
C3

Write a C program that declares an array of this structure to hold data for N days.

The program should use a loop to take input for all N days and then print out the weather records in a clean format.

b) Write a C program that reads n pH values (integers ranging from 0 to 14) from an input file named purity.txt.

4 CO3
C3

The program must evaluate each reading and write both the pH value and its classification code to an output file named report.txt using these rules:

- If pH < 6 → 'A' (Acidic)
- If pH > 8 → 'B' (Basic)
- Otherwise → 'N' (Neutral)

purity.txt File	report.txt File
3	5 A
5	9 B
9	7 N
7	

[OR]

Write a C program to read the names and marks of N students and write the information into a file named marks.txt. After closing the file, read the file contents and display them on the screen.

~~5~~ Determine the output of the following macro and bitwise operations. Show the calculation.

3 CO1
C3

- i) `#define MULTIPLY(a, b) a * b` ii) `int y = 12 >> 2;` iii) `int z = 22 << 2;`
`int x = MULTIPLY(3 + 2, 4);` `printf("%d\n", y);` `printf("%d\n", z);`
`printf("%d\n", x);`